

Where Does This IP Think It Is?

Geolocation Accuracy in the Age of Starlink, CGNAT, and Geofeeds

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Abstract

IP geolocation drives content delivery, fraud checks, and regional access decisions, yet an IP address does not always map to one place. Satellite terminals can receive addresses tied to distant points of presence, carrier-grade network address translation (CGNAT) puts subscribers from different regions behind one public address, and privacy relays keep only an approximate location by design. Geofeeds let operators publish where their prefixes are, but a 2024 study found them covering about 1.5% of allocated IPv4 prefixes. In our September 2026 crawl, 9.5% of 6,064 geofeed URLs could not be retrieved, and 534 entries across 80 feeds failed RFC 8805 validation. We separate three cases: wrong labels, which a geofeed can fix; subscribers sharing one address, which no feed can resolve; and deliberate coarsening by relays, which consumers should respect. We then give publishers and consumers a playbook and specify a versioned audit that tests retrieval, syntax, signatures, and geographic plausibility as separate properties.

Keywords

IP geolocation, geofeeds, satellite Internet, CGNAT, IPv6, privacy relays, measurement

1 Introduction

A Starlink terminal inside the Arctic Circle was listed as Vancouver in the operator’s geofeed, more than 2,400 km away [20]. A service that accepts this entry without verification inherits the same error. The harder question is what the published location was meant to describe: the terminal, the network point of presence (PoP) it attaches to, or the pool its address came from.

The report of the December 2025 IAB Workshop on IP Address Geolocation states the architectural problem plainly: IP addresses “are not designed to be indications of physical location” [13]. A registration address, a PoP, and a subscriber can all be tied to the same prefix, and a city field does not say which one it describes or how certain the publisher is.

This paper makes three contributions. First, it separates location failures a geofeed can fix from those it cannot and those it should leave alone, and places IPv6 and leased space in that frame (§3). Second, it reads the published geofeed measurements against the RFCs and turns the gaps into concrete steps for publishers and consumers (§4 and §5). Third, it specifies a reproducible audit and reports what our own crawl and measurement network already show (§6). We draw on recent measurement studies, the geofeed RFCs, and position papers from the IAB workshop, including our own [8, 19, 20, 25].

2 Background and the Accuracy Baseline

We build our location data in layers: registry records, operator-published geofeeds, user corrections, and active measurements such as ping and traceroute [26]. Registry data says who holds an address, not where it is used, and old location data often stays attached to blocks after they are sold and moved [13]. In 2011, Poese et al. concluded that databases “can claim country-level accuracy, but certainly not city-level” [22]. Fifteen years later, city-level accuracy depends on how and where a user connects.

Nabi et al. evaluated four providers using 37,302 observations from RIPE Atlas and UNICEF Giga across 175 countries, with 25.3% IPv6 observations and provider snapshots from March 2026 [18]. Median errors ranged from 179 to 207 km on mobile networks, compared with 3 to 16 km on fixed networks. With failure defined as error greater than 100 km, rates ranged from 53 to 61% in Asia, 66 to 72% in Africa, and 9 to 20% in Europe. They attribute the regional gap mainly to coarse provider prefixes; for mobile networks, prefix size explains only part of it. Fixed-network medians were much lower in this sample, although their 90th-percentile errors still exceeded hundreds of kilometers.

Ground truth has errors of its own. Izhikevich et al. found that at least 731 RIPE Atlas probes (2.4%) were likely not at their reported locations at some point between January 2019 and April 2026; the number rose from 22 in January 2019 to 155 in April 2026, with inferred errors often reaching hundreds or thousands of kilometers [14]. Any accuracy figure should state its date, reference source, sample, geographic unit, and error threshold. Agreeing with another database is not the same as being right.

3 Where Address-to-Place Mappings Break

Not every wrong location is a data error. Some labels are wrong and can be corrected. Some addresses are shared by subscribers in several places, and no label can fix that. Some services publish only a broad region on purpose. Which case applies decides who can fix the problem, if anyone can.

3.1 Satellite and roaming

Satellite terminals can receive addresses tied to a distant PoP. Starlink’s own position paper describes an aircraft moving from a PoP in Qatar to one in New York mid-flight while keeping the same address [19]. Pan and Zhao found the granularity of Starlink’s feed “highly inconsistent” and saw users stay mislocated for days or weeks after a correct feed update, because content providers kept using cached copies [20].

A conflicting estimate does not prove a feed is stale. APNIC’s ad-based user estimates once assigned about 6 million users in Yemen to Starlink, and APNIC overrode Starlink’s geofeed for 20 economies [12]. Eight weeks later APNIC restored the feed, after tracing the Yemen figure to a 90% drop in ad volume on Yemen Net, the country’s main ISP [11]. The second look gave APNIC “a lot more confidence in the fidelity of the Starlink geolocation data.” Ships at sea and aircraft in flight remained an open question.

3.2 Carrier-grade NAT

CGNAT can put “hundreds or even thousands” of clients behind one public IPv4 address [9]. When those clients are in different regions, no single location fits the address, and a more-specific prefix cannot help because they already share it. A geofeed can name the region the pool serves; it cannot separate subscribers the address has already merged (Figure 1). The cost falls unevenly: Giotsas and Fayed note that actions on a shared address “may disproportionately affect populations in developing nations” [9].

3.3 Privacy relays

Privacy relays coarsen location on purpose, and not all in the same way. Apple says iCloud Private Relay “preserves the region the user is in” and publishes its egress ranges with their locations [1]. Cloudflare says WARP gives users an address that represents their “approximate location” [3] and “does not allow you to pretend to be accessing the Internet from a different country” [4]. Some VPNs and proxies instead deliberately change how the user appears to geolocation providers [13]. A relay address therefore says little about intent. Distinguish the user’s coarse region from the relay exit’s location, and make access decisions separately. Being a relay address is not evidence of abuse.

3.4 IPv6

IPv6 can remove the scarcity-driven need for subscribers to share a public address. It does not make addresses geographic. A global unicast IPv6 address contains a routing prefix, a subnet ID, and an interface ID [10]; none is a geographic field. In a direct IPv6 connection, ordinary routing preserves the client’s source address rather than replacing it with an egress router’s address [5]. Location inference can still fail when assigned prefixes span several places, mappings lag renumbering, or a proxy substitutes its own address. More address space permits finer assignments; accurate geolocation still needs matching, maintained location data. Livadariu et al.’s 2024 study found geofeed coverage for only 0.70% of allocated IPv6 prefixes, against 1.50% for IPv4 [16]. EUI-64 identifiers raise a separate privacy issue, covered in Appendix A.

3.5 Coarse and leased blocks

A large prefix can hold subprefixes in different cities, and RFC 8805 lets a publisher list each one: nested entries are allowed and consumers use the longest match [15]. Broad aggregation is a publishing choice, not a format limit (Figure 1). When one address serves several places, publish only geographic fields common to the served area and leave unsupported fields blank; every field after the prefix is optional [15]. Filling them anyway creates false precision:

Gasser et al. found Starlink’s feed listing each country’s capital as the city [8].

Leasing adds conflicts. When a lessor and a lessee both publish the same range, consumers see two cities for one prefix. Our earlier workshop paper proposed that the lessee publish the subprefix and the lessor remove its conflicting entry for that subprefix [25]. Updates also have to follow relocations, reassignments, lease returns, and PoP changes. Watching only new allocations misses them.

4 Geofeed Deployment and Its Limits

RFC 8805 defines the CSV format, RFC 9632 defines discovery through RPSL inetnum objects and optional RPKI signatures, and RFC 9877 adds geofeed links to RDAP [2, 15, 24]. Whether a file can be fetched, whether it parses, whether the address holder signed it, and whether its locations are plausible are four separate questions.

Coverage. Fainchtein and Sherr observed a tenfold increase in the IPv4 addresses covered by geofeeds during August 2022 to October 2023, but coverage remained below 1% of the IPv4 address space [7]. Livadariu et al.’s 2024 study reported coverage of about 1.50% of allocated IPv4 prefixes and 0.70% of allocated IPv6 prefixes [16]. One study counts addresses and the other prefixes, but both put coverage under 2%. Adoption also skews toward more industrialized nations [7], while Nabi et al. found the largest errors in Asia and Africa in their sample [18].

Conformance and retrieval. In a February to March 2024 crawl, Desai et al. could fetch 1,427 of 1,547 discovered geofeed URLs (7.76% failed) and classified 10.49% of 570,909 lines as malformed under their validation rules [6]. Gasser et al. show what such errors look like: non-ISO region codes, swapped country and region fields, and an IPv6 prefix typed as /6 instead of /64 [8]. RFC 8805 treats duplicate prefixes as an error but allows nested ones [15], so a validator has to tell the two apart.

Maintenance. An RFC 8805 entry has no timestamp or expiry. Without an explicit refresh signal, RFC 8805 recommends refreshing at least weekly and RFC 9632 recommends fetching at most weekly [2, 15]. Together, they imply a weekly default. That is slow for Starlink, which holds unused addresses for up to a month to cover slow updates and asks for a 24-hour cadence [19]. A changed file shows that the publisher edited it, and an unchanged file can still be right, so content-change times measure publishing behavior, not location accuracy.

Expressiveness. A row cannot say how confident the publisher is, whether a location belongs to the user or the infrastructure, or that one address serves subscribers in several places. The IAB report lists these among the improvements geofeeds need [13]. Most entries Livadariu et al. could check looked correct; erroneous geofeed locations affected 0.9% of client, 4.0% of router, and 8.5% of server addresses [16].

Authentication. RFC 9632 lets the address holder sign a feed: a detached Cryptographic Message Syntax (CMS) signature over canonicalized feed content, made with an RPKI end-entity certificate that covers every prefix in it and carries no AS numbers or inherited resources. A consumer checks that the certificate is on the current RPKI manifest and validates its chain, then verifies the

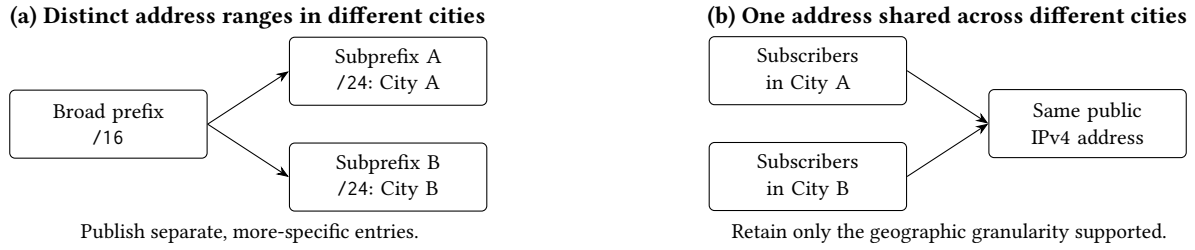


Figure 1: Conceptual examples of prefix granularity and shared-address ambiguity. RFC 8805 permits nested entries and longest-prefix matching [15]. That mechanism can distinguish address ranges within a larger block; it cannot distinguish subscribers who simultaneously share one public address.

signature and address coverage [2]. A valid signature shows that the holder of those addresses signed the file and that it has not changed since. It says nothing about whether the locations are right. Signing is optional, and RFC 9632 requires consumers of unsigned feeds to ignore entries outside the address range of the record that pointed to the feed.

5 An Operator Playbook

None of this is exotic. Each step below targets a failure described in §3 or §4, and the audit in §6 measures how consistently publishers follow them.

For publishers. Generate feeds from maintained allocation and deployment records. Trigger updates on physical relocation, subscriber or pool reassignment, lease changes and returns, and changes in the geographic area served by a PoP. Check the resulting location claim even when a routing change leaves the address allocation unchanged.

Validate the IP prefix and each populated location field before publication; detect exact duplicates while allowing legitimate nested prefixes. Publish more-specific entries when distinct ranges have distinct locations. Use the coarsest accurate value for shared pools, and leave unsupported optional fields blank. Serve the file at a stable HTTPS URL, publish the discovery pointer, and set HTTP cache headers to match the expected update cadence. Coordinate lessor and lessee records and sign the feed when RPKI signing is available [2, 8, 15, 24, 25].

For consumers. Treat IP geolocation as coarse evidence, and match each decision to the precision the data supports. Keep each feed’s discovery record, scope, fetch time, and signature result. Apply the RFC 9632 selection rules for discovery objects and authenticated sources, followed by RFC 8805 longest-prefix matching within a selected feed. Record unresolved equally specific conflicts for investigation rather than silently depending on input order [2, 15].

Keep parsing errors separate from geographic disagreements. A valid, signed row can still be wrong; a failed probe leaves a location untested. Check responsive addresses from several vantage points, keep an indeterminate result when evidence is thin, and read relay entries at their stated granularity. Use path-appropriate propagation bounds: Izhikevich et al. exclude Starlink because satellite links can beat a terrestrial-fiber bound [14]. There is also no standard way to report a bad entry: Gasser et al. describe consumers ignoring the entry, guessing the intended value, or emailing the WHOIS

contact [8], and the IAB report notes that ISPs have no standardized way to correct third-party databases [13]. Keep the evidence for each dispute so the fix can be checked when it lands.

The IP Geolocation HTTP Client Hint, an expired individual Internet-Draft, proposes carrying a coarse IP-derived location from the client when the server opts in [21]. The client obtains it from a cooperating server’s IP database lookup, not GPS, and it must be no more precise than the IP reveals. The draft forbids its use for security or access control. It changes how a location claim is conveyed; it does not provide independent ground truth.

6 Toward Verifiable Geofeeds

Operational observations. Our production crawlers read registry data and geofeeds continuously [26]. In September 2026 they requested 6,064 geofeed URLs found in public WHOIS records; 9.5% could not be retrieved, and our validator flagged 534 entries across 80 feeds as nonconformant with RFC 8805.

We check feed claims with IPGeolocation.io’s Vantage Network, whose measurement points ping and traceroute target addresses. Round-trip times from known locations constrain plausible target locations [26]. When several vantage points agree that a prefix is not where its feed says, we lower the feed’s weight for that prefix and investigate it. The methodology post illustrates four discrepancies of 396 to 9,496 km between feed claims and inferred response locations, including Mexico City versus Frankfurt [26]. Separately, one operational case listed a /24 in Canada while reverse-DNS hints and latency measurements indicated Germany. Coverage is the hard limit: in the methodology post’s full scan, only 10.05% of public, unreserved IPv4 addresses answered at least one probe, with a bias toward routers and servers [26].

A reproducible audit. The audit we propose follows each claim from discovery to a verdict. It collects RPSL geofeed pointers and RDAP links with their address scopes, deduplicates URLs under a published rule, fetches on a fixed schedule, and preserves raw bytes, response metadata, content hashes, and parser and validator versions. Each stage reports its own denominator. Retrieval, conformance, and plausibility checks already run in our pipeline; the audit adds a fixed schedule, published denominators, and preserved snapshots so that others can reproduce the numbers. Mok et al.’s ongoing longitudinal study collects feeds daily to examine publication and format compliance [17]; our audit also makes authentication and network plausibility explicit outcomes.

The audit reports six metrics:

- **Coverage:** routed prefixes with discoverable, in-scope geofeed coverage, separating full from partial coverage and IPv4 from IPv6. Report the routing snapshot and an address-weighted measure alongside prefix counts.
- **Retrieval:** successful file retrievals per discovered URL, with DNS, timeout, HTTP, and non-feed responses classified separately. Publish the timeout and retry policy.
- **Conformance:** valid and invalid data rows per retrieved feed, with rule-specific error counts. Keep comments, exact duplicates, and permitted nested prefixes distinct.
- **Observed maintenance:** first-seen, last-seen, and last content-change times, plus publisher timestamps where available. These measure how publishers behave, not whether locations are current.
- **Authentication:** valid, invalid, and absent signatures, with the RPKI snapshot and validation time recorded.
- **Plausibility:** agreement, contradiction, or insufficient evidence against latency and traceroute measurements. Report target selection, vantage locations, measurement times, filtering rules, and distance or latency thresholds. Probe failure is not a geographic error.

These records let another group tell a changed feed from a changed validator. A year of monthly snapshots could track retrieval and conformance failures as adoption changes, and bound correction delays and disputed-entry persistence at the sampling resolution. A study with consented, location-tagged satellite terminals could test how well feed claims describe roaming users.

7 Conclusion

A geofeed can fix a wrong label and sharpen a broad one. It cannot separate subscribers who share one address, and it should not undo the coarsening a privacy relay applies on purpose. IPv6 eases address sharing but does not make addresses geographic. The near-term work is concrete: publish feeds that are generated from source records, validated, signed, and accurate about their granularity; read them as coarse evidence and check them against the network; and run a shared audit so the community can see whether failures like the 9.5% retrieval failure rate in our crawl are going down.

References

- [1] Apple Inc. n.d. Prepare Your Network or Web Server for iCloud Private Relay. Apple Developer Documentation. Accessed 5 October 2026. <https://developer.apple.com/icloud/prepare-your-network-for-icloud-private-relay/>.
- [2] Randy Bush, Massimo Candela, Warren Kumari, and Russ Housley. 2024. Finding and Using Geofeed Data. RFC 9632, IETF. Obsoletes RFC 9092. <https://www.rfc-editor.org/rfc/rfc9632.html>.
- [3] Cloudflare, Inc. 2026. WARP Client: Frequently Asked Questions. Cloudflare Developer Documentation. <https://developers.cloudflare.com/warp-client/know-issues-and-faq/>.
- [4] Cloudflare, Inc. 2026. WARP Modes. Cloudflare Developer Documentation. <https://developers.cloudflare.com/warp-client/warp-modes/>.
- [5] Stephen Deering and Robert Hinden. 2017. Internet Protocol, Version 6 (IPv6) Specification. RFC 8200, IETF. <https://www.rfc-editor.org/rfc/rfc8200.html>.
- [6] Dipsy Desai, Kicho Yu, and Sulyab Thottungal Valapu. 2025. Geofeed Adoption and Authentication. In *Proc. IEEE/IFIP Network Operations and Management Symposium (NOMS)*, 1–8. <https://doi.org/10.1109/NOMS57970.2025.11073659>.
- [7] Rahel A. Fainchtein and Micah Sherr. 2024. You Can Find Me Here: A Study of the Early Adoption of Geofeeds. In *Proc. Passive and Active Measurement (PAM) (LNCS, Vol. 14538)*. Springer, 228–245. https://doi.org/10.1007/978-3-031-56252-5_11.
- [8] Oliver Gasser, William Leung, and Maxime Mouchet. 2025. Challenges of Working With Geofeeds. Position paper, IAB Workshop on IP Address Geolocation. <https://datatracker.ietf.org/doc/slides-ipgeows-paper-challenges-of-working-with-geofeeds/>.
- [9] Vasilis Giotas and Marwan Fayed. 2025. One IP Address, Many Users: Detecting CGNAT to Reduce Collateral Effects. The Cloudflare Blog. Published 29 October 2025. <https://blog.cloudflare.com/detecting-cgn-to-reduce-collateral-damage/>.
- [10] Robert Hinden and Stephen Deering. 2006. IP Version 6 Addressing Architecture. RFC 4291, IETF. <https://www.rfc-editor.org/rfc/rfc4291.html>.
- [11] Geoff Huston. 2025. A Second Look at Geolocation and Starlink. APNIC Blog. Published 25 November 2025. <https://blog.apnic.net/2025/11/25/a-second-look-at-geolocation-and-starlink/>.
- [12] Geoff Huston. 2025. Geolocation and Starlink. APNIC Blog. Published 30 September 2025. <https://blog.apnic.net/2025/09/30/geolocation-and-starlink/>.
- [13] Jana Iyengar, Jason Livingood, and Tommy Pauly. 2026. Report from the IAB Workshop on IP Address Geolocation. Internet-Draft draft-iab-ip-geo-workshop-report-04. Internet Architecture Board. Work in Progress. <https://datatracker.ietf.org/doc/draft-iab-ip-geo-workshop-report/>.
- [14] Katherine Izhikevich, Ben Du, Manda Tran, Sumanth Rao, Alisha Ukani, Ray Bellis, and Liz Izhikevich. 2026. Rooting Out Incorrect RIPE Atlas Probe Geolocations. arXiv:2409.19109v4 [cs.NI]. To appear in Proc. ACM Internet Measurement Conference (IMC 2026). <https://arxiv.org/abs/2409.19109v4>.
- [15] Erik Kline, Krzysztof Duleba, Zoltan Szamonek, Stefan Moser, and Warren Kumari. 2020. A Format for Self-Published IP Geolocation Feeds. RFC 8805, RFC Editor (Independent Submission). <https://www.rfc-editor.org/rfc/rfc8805.html>.
- [16] Ioana Livadariu, Kevin Vermeulen, Maxime Mouchet, and Vasilis Giotas. 2024. Geofeeds: Revolutionizing IP Geolocation or Illusory Promises? *Proceedings of the ACM on Networking* 2, CoNEXT3, Article 15 (2024), 21 pages. <https://doi.org/10.1145/3676869>.
- [17] Ricky Mok, Parwiz Khan, Eduard Dehelean, and Rocky Chang. 2026. Enabling a Longitudinal Study of Geofeed Data Quality, Adoption, and Dynamics. AIMS-19 workshop presentation, 25 February 2026. https://www.caida.org/catalog/media/2026_enabling_longitudinal_geofeed_aims/enabling_longitudinal_geofeed_aims.pdf.
- [18] Syed Tauhidun Nabi, Jocelyn Bliton, Tijay Chung, and Shaddi Hasan. 2026. Lost in the Prefix: Revisiting IP Geolocation Accuracy Across Networks and Geographies. arXiv:2605.21937v1 [cs.NI]. <https://arxiv.org/abs/2605.21937v1>.
- [19] Nathan Owens. 2025. Position Paper: Starlink. Position paper, IAB Workshop on IP Address Geolocation. <https://datatracker.ietf.org/doc/slides-ipgeows-paper-position-paper-starlink/>.
- [20] Jianping Pan and Jinwei Zhao. 2025. GeoFeed in the Wild: A Case Study on StarlinkISP.net. Position paper, IAB Workshop on IP Address Geolocation. <https://datatracker.ietf.org/doc/slides-ipgeows-paper-geofeed-in-the-wild-a-case-study-on-starlinkispnet/>.
- [21] Tommy Pauly, David Schinazi, Ciara McMullin, and Dustin Mitchell. 2025. The IP Geolocation HTTP Client Hint. Internet-Draft draft-pauly-httpbis-geoip-hint-02. IETF. Work in Progress; individual submission, expired April 2026. Also a position paper at the IAB Workshop on IP Address Geolocation. <https://datatracker.ietf.org/doc/draft-pauly-httpbis-geoip-hint/>.
- [22] Ingmar Poese, Steve Uhlig, Mohamed Ali Kaafar, Benoit Donnet, and Bamba Gueye. 2011. IP Geolocation Databases: Unreliable? *ACM SIGCOMM Computer Communication Review* 41, 2, 53–56. <https://doi.org/10.1145/1971162.1971171>.
- [23] Erik C. Rye and Robert Beverly. 2023. IPSeeYou: Exploiting Leaked Identifiers in IPv6 for Street-Level Geolocation. In *Proc. IEEE Symposium on Security and Privacy (S&P)*. IEEE, 3129–3145. <https://doi.org/10.1109/SP46215.2023.10179376>.
- [24] Jasdeep Singh and Tom Harrison. 2025. Registration Data Access Protocol (RDAP) Extension for Geofeed Data. RFC 9877, IETF. <https://www.rfc-editor.org/rfc/rfc9877.html>.
- [25] Mudassar Tariq. 2025. IP Address Geolocation: Use Cases, Gaps, and Future Directions. Position paper, IAB Workshop on IP Address Geolocation. <https://datatracker.ietf.org/doc/slides-ipgeows-paper-ip-address-geolocation-use-cases-gaps-and-future-directions/>.
- [26] Mudassar Tariq. 2026. IP Geolocation Data: Built, Not Guessed. IPGeolocation.io. Published 13 July 2026. <https://ipgeolocation.io/blog/how-ip-geolocation-data-is-built>.

A Ethics and Privacy

Ethics. Besides published studies, this paper uses two kinds of our own data: a crawl of geofeed files that operators publish for consumers, and ping and traceroute measurements from the Vantage Network [26]. The measurements use round-trip times and traceroute paths to check claims about network addresses. They involve no end-user traffic and do not identify individual subscribers. A future study with satellite terminals would need consent from

terminal owners and should publish only the precision its question needs.

Privacy. Better coarse geolocation and precise location leakage are separate problems. IPvSeeYou geolocated the IPv6 prefixes of more than 12 million routers in 146 countries and territories, with

a 39 m median error on its validation set, by inferring Wi-Fi MAC addresses from the WAN MAC addresses embedded in EUI-64 interface identifiers and looking them up in wardriving databases [23]. That leak comes from how some devices form addresses, not from geolocation databases, and coarse geolocation should never rely on it.